17 In this question you must show detailed reasoning

Jars of honey contain a nominal mass of 340 grams of honey.

The jars are filled by an automated process so that the actual mass of honey in a jar, X grams, is Normally distributed with mean 350 grams. This is to minimize the chance of a jar which contains less than 340 grams of honey being sold.

During a quality control check, a random sample of 50 jars of honey are checked.

Summary statistics for the check are as follows.

$\Sigma x = 17367 \quad \Sigma x^2 = 6036597 \quad n = 50$

Carry out a hypothesis test at the 1% level to see if there is any evidence to suggest that the mean mass of honey in the jars is less than 350 grams. **[9]**

17

(answer space continued on next page)

17 **(continued)**